

ENVS 193DS Spring Final

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Set Up

```
# reading in packages
library(tidyverse)
library(janitor)
library(DHARMA)
library(here)

# reading in nest data
nest <- read.csv(here("data/nest_data_final.csv"))
# reading in my personal data
my_data <- read.csv(here("data/personal_data.csv"))
```

Problem 1. Research writing

a. Transparent statistical methods

In part one my co-workers are using a generalized linear model test. This is seen through the response variable, whether the American Avocet utilized the habitat (San Francisco Bay), being binary data (yes or no). Additionally the binary response variable is being linked to a continuous variable, the distance from the water edge, further signalling the use of a generalized linear model.

In part two my co-workers are using a chi-squared test. This is seen through the analysis testing whether there is a relationship between two categorical variables: bird species and habitat.

b. Suggestions for rewriting

- **Part 1:** The data indicates that the distance to the water's edge significantly influences whether American Avocets utilizes a particular microhabitat. This means an avocet's likelihood of occupying an area changes predictably depending on how close or far the area is from the edge of the water.
- **Part 2:** We found clear evidence that different shorebird species (American Avocet, Forter's Tern, Black-necked Stilt) prefer distinct types of habitats (managed pond, non-tidal marsh, salt pond, tidal marsh, or tidal mudflat) over others in the San Francisco Bay.

c. Further analysis

- **Additional Variable:** An additional variable that could influence American Avocet microhabitat use is water depth.
- **Variable Type:** Water depth is a continuous variable that can be measured in centimeters (cm).
- **Why?:** Water depth may influence habitat use because American Avocets are wading shorebirds that require shallow water to effectively find food. Due to this, analyzing water depth could compliment the distance to water edge data and provide a more comprehensive understanding to the specific ecosystem conditions that impact shorebird habitat use.

Problem 2. Data analysis

a. Clean and wrangle

```
# creating clean object
nests_clean <- nest |>
# cleaning names
clean_names() |>
# selecting for desired columns
select(height_cm,
        case_control,
        ant_species) |>
# filtering for multiple values using %in%
filter(ant_species %in% c("AB", "RRB", "BBR")) |>
# full scientific names in new column
mutate(species_name = fct_recode(as.factor(ant_species),
```

```

    "C. sjostedti" = "AB",
    "C. mimosae"   = "RRB",
    "C. nigriceps" = "BBR"
  )) |>
  # reordering columns in desired order
  mutate(species_name = fct_relevel(species_name,
                                    "C. sjostedti",
                                    "C. mimosae",
                                    "C. nigriceps")) |>

  # arranging by species_name
  arrange(species_name)
# displaying 10 random rows
slice_sample(nests_clean, n = 10)

```

	height_cm	case_control	ant_species	species_name
1	155.5	0	RRB	C. mimosae
2	237.5	0	RRB	C. mimosae
3	410.5	1	RRB	C. mimosae
4	180.5	1	RRB	C. mimosae
5	146.5	0	RRB	C. mimosae
6	238.5	0	RRB	C. mimosae
7	166.5	0	RRB	C. mimosae
8	155.5	0	RRB	C. mimosae
9	234.5	0	RRB	C. mimosae
10	249.0	0	RRB	C. mimosae

b. Response variable

c. Purpose of study

d. Table of models

e. Run the models

Select the best model

g. Check the diagnostics

h. Visualize the model predictions

i. Write a caption for your figure

j. Calculate model predictions

k. Interpret your results